

8	(a) op-amp has infinite/(very) large gain	B1	
	op-amp saturates if $V^+ \neq V^-$	M1	
	V^+ is at earth potential so P (or V^-) must be at earth	A1	[3]
	(b) input resistance to op-amp is very large or current in R_2 = current in R_1	B1	
	$V_{IN} (-0) = IR_2$ <u>and</u> $(0) - V_{OUT} = IR_1$	M1	
	$V_{OUT} / V_{IN} = -R_1 / R_2$	A1	[3]
	(c) relay coil connected between V_{OUT} and earth	M1	
	correct diode symbol connected between V_{OUT} and coil or between coil and earth	M1	
	correct polarity for diode ('clockwise')	A1	[3]